

Impacts of urbanization on urban structures and energy demand: What can we learn for urban energy planning and urbanization management?

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ABSTRACT

Since 2007, for the first time in human history, more than half of the world's population has been living in cities. The urbanization process is a key phenomenon of economic development, and leads to a significant concentration of human resources, economic activities, and resource consumption in cities. Although covering only about 2% of the earth's surface, cities are responsible for about 75% of the world's consumption of resources. This trend will intensify over the next decades as a consequence of high urbanization rates in Africa and, even more importantly, in Asia. In order to estimate the impact of urbanization on energy demand, we have to identify the different processes and mechanisms of urbanization that substantially affect urban structures as well as human behavior. Taking a closer look at city-related production, mobility and transport, infrastructure and urban density, as well as private households, we find that various mechanisms of urbanization within the different sectors of the economy lead to a substantial increase in urban energy demand and to a change in the fuel mix. The relevance of these mechanisms differs considerably between developed and developing countries as well as within the group of developing countries. Over the next decades, cities and especially newly emerging megacities in developing countries will play a key role concerning the development and distribution of global energy demand. Hence, urban energy planning and urbanization management will be pivotal for creating the right framework conditions for a sustainable energy future.

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1. Introduction

Since 2007, for the first time in human history, more than half of the world's population has been living in cities. Urbanization and industrialization characterized the economic development in Europe, Northern America, and Japan already during the 19th and early 20th century, and led to a continuous economic, demographic, functional, and extensive growth of cities (Bähr & Jürgens, 2005). A comparable development has not yet appeared in most developing countries. Under conditions of colonialism, economic development was limited to a rudimentary form of industrialization that also determined the design, infrastructure, and function of cities in most developing countries. Hence, an increasing degree of urbanization in developing countries became an issue only in the second half of the 20th century, when most countries in Africa and Asia regained their independence. Today, developing countries in Africa and Asia show an average degree of urbanization that industrial countries were experiencing between 1900 and 1925 (UNPD, 2008). Table 1

shows the development and prospects of urbanization for the world and major regions.

Comparing the evolution of the population in urban areas between 1970 and 2010 in more and less developed regions,¹ it becomes clear that urban growth has mainly occurred in less developed countries. This trend seems to continue. The urban population in less developed countries is expected to nearly double within the next 40 years, from about 2.6 to 5.3 billion people. In doing so, least developed countries are expected to have the highest average urban growth rate of 3.3% per annum between 2010 and 2050. According to the projections of the United Nations, about 83% of the world's urban population in 2050 will live in less developed regions. Within the group of less developed countries, African and Asian cities will experience higher urban growth rate prospects over the next 40 years. In contrast, urban growth in more developed regions, such as North America and Europe, is almost saturated and expected to reach zero within the next 40 years. Fig. 1 reveals the future role of less developed regions with regard to urban population growth

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¹ We apply the United Nations definition of more and less developed regions and least developed countries (UN World Population Prospects Database, <http://esa.un.org/unpp/definition.html>).

Table 1
Urbanization development and prospects, worldwide and in major regions, 1970–2050.

	Population in urban areas (millions)			Urban growth rate (%)		Urban share (%)
	1970	2010	2050	1970–2010	2010–2050	
World	1332	3495	6398	2.4	1.5	50.6
More developed regions	652	925	1071	0.9	0.4	75
Less developed regions	680	2570	5327	3.3	1.8	45.3
Least developed countries	41	254	967	4.5	3.3	29.4
Africa	86	412	1234	3.9	2.7	39.9
Asia	485	1770	3486	3.2	1.7	42.5
Latin America/Caribbean	164	471	683	2.6	0.9	79.4
Northern America	171	286	401	1.3	0.9	82.1
Europe	412	530	557	0.6	0.1	72.6
Oceania	14	25	37	1.5	1	70.6

Source: UNPD (2007), own calculation.

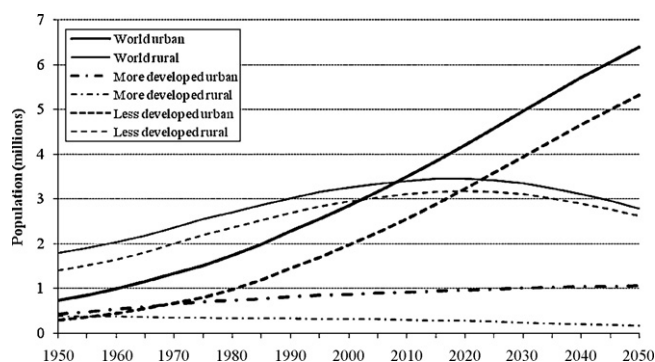


Fig. 1. Urban and rural population growth for the whole world (solid lines), the more developed (hashed lines), and the less developed (dashed lines) regions, 1950–2050. Source: UNPD (2007), own illustration.

and the importance of cities. The illustration provides some evidence that the worldwide urban growth is mainly driven by less developed countries, which becomes clear from the parallel evolution of the world's and the less developed countries' urban and rural populations.

With regard to economic development, future urbanization will be one of the key processes particularly in the developing world, where countries are currently facing an extensive quantitative growth of cities that impinges on urban structures and energy demand.

Covering only 2% of the world's surface, cities are responsible for about 75% of the world's consumption of resources (Pacione, 2009). Table 2 shows world energy demand in cities by fuel. It can be seen that the world's energy demand is mainly characterized by urban demand. Two thirds of the world's total energy consump-

tion of 7908 Mtoe and 70% of the CO₂ emissions are attributable to cities. With regard to rapidly growing cities in developing countries over the coming years, the International Energy Agency predicts an increase in the share of urban energy demand to 73% and of CO₂ emissions to 76% by 2030 (IEA, 2008). Moreover, the worldwide urban energy demand is dominated by fossil fuels. The global rise and regional spread of most large cities today is founded on the availability of high-energy density, based on centralized and inexpensive fuels, such as coal, oil and natural gas (Droege, 2004). Coal, oil, and natural gas will also exhibit the largest shares in the next decades. Newly industrializing countries, such as China, will reinforce this trend (China, for example, currently represents 20% of the world's population and 17% of the world's urban population, but only features a degree of urbanization of 40%). Already today, 75% of energy in China is consumed in cities (by 2030 projected to have risen to 83%; cf. IEA, 2008).

Comparing large cities in developed and developing countries, it can be found that average energy consumption per capita in developing countries is less than in developed countries (Fig. 2).

On average, most energy is consumed in Northern American cities (51,000 MJ), followed by cities in Oceania (29,500 MJ) and Western Europe (16,000 MJ), respectively. African and less developed Asian cities consume least of all. This might be due to the fact that the consumption of non-commercial/traditional energy sources is not included in these figures. Under conditions of lacking access to electricity, traditional energy sources, such as wood fuel, crop residues, and animal dung for cooking, lighting, and space heating, play a key role with regard to the energy supply of the poor not only in rural areas (Madlener, 2009, p. 741). If the energy consumption of cities in Asia or Africa increases to the level of the energy consumption of Northern American cities under today's conditions (e.g. with regard to the fuel mix), an ecological collapse will inevitably be the consequence (Gaebe, 2004).

Table 2
World energy demand in cities by fuel in the IEA reference scenario.

	2006		2015		2030		2006–2030 ^a
	Mtoe	Cities as a % of world	Mtoe	Cities as a % of world	Mtoe	Cities as a % of world	
Coal	2330	76	3145	78	3964	81	2.2
Oil	2519	63	2873	63	3394	66	1.2
Natural Gas	1984	82	2418	83	3176	87	2.0
Nuclear	551	76	630	77	726	81	1.2
Hydro	195	75	245	76	330	79	2.2
Biomass and wastes	280	24	358	26	520	31	2.6
Other renewables	48	72	115	73	264	75	7.4
Total	7908	67	9785	69	12374	73	1.9
Electricity	1019	76	1367	77	1912	79	2.7

Source: IEA (2008), own illustration.

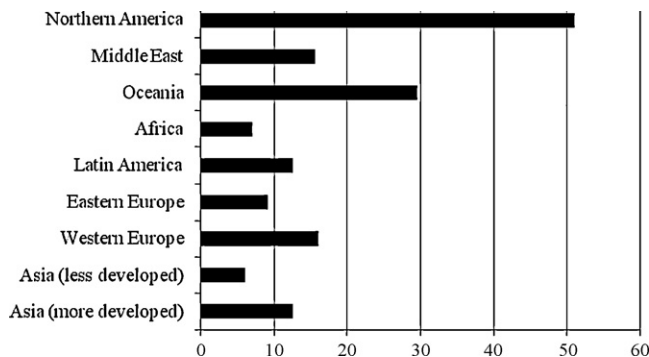
^a Average annual growth rate.

Table 3

Population and the world's megacities, 2007–2025 (in millions).

Rank	2007		Projection for 2025	
	City	Population	City	Population
1	Tokyo	35.7	Tokyo	36.4
2	New York–Newark	19.0	Mumbai (Bombay)	26.4
3	Mexico City	19.0	Delhi	22.5
4	Mumbai (Bombay)	19.0	Dhaka	22.0
5	São Paulo	18.8	São Paulo	21.4
6	Delhi	15.9	Mexico City	21.0
7	Shanghai	15.0	New York–Newark	20.6
8	Kolkata (Calcutta)	14.8	Kolkata (Calcutta)	20.6
9	Dhaka	13.5	Shanghai	19.4
10	Buenos Aires	12.8	Karachi	19.1
11	Los Angeles	12.5	Kinshasa	16.8
12	Karachi	12.1	Lagos	15.8
13	Cairo	11.9	Cairo	15.6
14	Rio de Janeiro	11.7	Manila	14.8
15	Osaka–Kobe	11.3	Beijing	14.5
16	Beijing	11.1	Buenos Aires	13.8
17	Manila	11.1	Los Angeles	13.7
18	Moscow	10.5	Rio de Janeiro	13.4
19	Istanbul	10.1	Jakarta	12.4
20			Istanbul	12.1
21			Guangzhou	11.8
22			Osaka–Kobe	11.4
23			Moscow	10.5
24			Lahore	10.5
25			Shenzhen	10.2
26			Chennai (Madras)	10.1
27			Paris	10.0

Source: UNPD (2008), own illustration.

**Fig. 2.** Average energy consumption of large cities by regions, 1995, in 1000 MJ.

Notes: The numbers provided in this figure are based on a study that compared 100 large cities worldwide. Specifically, the study considered 15 cities in North America, 10 in Latin America, 35 in Western Europe, 6 in Eastern Europe, 8 in Africa, and 26 in Asia and Oceania.

Source: Gaebel (2004), own illustration.

Table 3 shows the world's emerging megacities with more than 10 million inhabitants. According to the definition of the United Nations, a city with more than 8 million inhabitants is a megacity, whereas the threshold value in the literature ranges from 5 to 10 million (Bähr & Jürgens, 2005). In 2007, worldwide 19 cities had more than 10 million inhabitants, 14 of which were located in developing countries. By 2025, the number of megacities is expected to have increased to 27, 22 of which will be located in developing countries. The trend of a high degree of urban primacy in many developing countries will intensify, with increasing shares of urban population concentrated in large cities (Bertinelli & Black, 2004). Also associated with the megacity issue is the decreasing accountability of cities due to extensive quantitative growth.

2. Impacts of urbanization on urban structures and energy demand

In order to estimate the impact of urbanization on energy demand, we analyze the city sectors, identifying the different processes and mechanisms of urbanization that substantially affect urban structures as well as energy consumer needs and behavior.

2.1. Urban production

Apart from population concentration, urbanization also concentrates economic activities in the city. Therefore, rural–urban migration accompanies structural transformation of the economy, because by moving into the city, the labor force is transferred from the agricultural sector in the rural areas to the industrial and service sectors in urban areas. The structural transformation of the economy causes a change in energy consumption, too, because the production shifts from low-energy intensity agricultural production to the production of high-energy intensity, specialized commodities, such as metals or chemicals. Although the transformation of production (and with it the change in energy consumption) is affected by the introduction of new technologies and industrialization, urbanization is also a major factor. Urbanization concentrates population in cities and, therefore, creates the potential for economic development. Likewise, the volume of production and the market range increase as a consequence of growing urbanization rates. This also leads to increasing transport distances that requires more transport energy (Jones, 1989, 1991, 2004; Parikh & Shukla, 1995).

Due to increasing urbanization, industrialization and not least tertiarization,² the size of the labor force in agricultural production decreases. This leads to a decreasing share of producers of

² Tertiarization implies the shift of labor and value added from the agricultural and also from the industrial sector to the service sector.

agricultural commodities compared to its consumers. As a consequence, on the one hand, many commodities have to be imported and transported over long distances while, on the other hand, agricultural production has to be intensified and mechanized (Jones, 2004). Ultimately, both developments lead to an increasing energy demand.

The supply of an increasing urban population, increasing competitive pressure due to economic development, and rising scarcity of available land require innovation in production and, associated with it, the substitution of traditional energy sources, and the introduction of modern, flexible, and reliable energy sources (Jones, 1989).

For the case of most developing countries, the measurement of energy consumption of urban production affected by urbanization is difficult due to the role of informal markets. In informal markets, or the informal sector, economic activity is neither registered nor taxed by the government and, therefore, not included in the GDP. In India, for example, it has been estimated that some 93% of the incomes and about 64% of the financial savings arise from informal activities (Hall & Pfeiffer, 2000).

2.2. Mobility and transport

As mentioned above, the concentration of population and economic activity generates new demand for transport services and supplies. Cities as nodal points of production rely on resource supplies that imply transport over long distances. Increasing the concentration of production and the labor force in urban areas raises transport needs and, as a consequence, also the consumption of fossil fuels. In most developed countries, city logistic concepts are applied in order to generate reliable and efficient supply and transport systems. In view of lacking basic infrastructure, many developing countries, especially in Sub-Saharan Africa, face other problems as well. A provisional supply of poorer city quarters with basic services, such as water or gas, must often be effected via truck transport (Pacione, 2009). Associated with a congested traffic, it is practically impossible to apply city logistic concepts in the way they are applied in more developed countries.

A major factor affecting urban energy demand in both developed and developing countries is motorized individual transport. Under conditions of continuous city growth and rural–urban migration, private transport is increasing substantially. Urbanization increases also inner-city private transport, because of commuter traffic, often over great distances (Jones, 2004). This tends to result in an increasing level of motorized individual transport that implies increasing energy consumption and emissions. Mobility represents a share of 25–60% of household energy consumption (Pacione, 2009), if

accounted under this heading. Table 4 shows modal shares of urban transport trips in Asian megacities in 2005 as an illustration.

The highly developed megacity of Tokyo shows the highest public transit share (53%) of all megacity regions due to a fully developed and highly efficient public transport system. Indian megacities, such as Kolkata or Delhi, have the highest shares of paratransit (40% and 53%, respectively) and only very modest shares of public transit (6% and 8%, respectively). This can be attributed to an insufficiently developed public transport system, but also partly to poorly developed street networks and congested traffic. The chaotic traffic conditions in many overcrowded and still growing megacities in developing countries with permanent congestion, and a lack of exhaust emission control for vehicles, have strong impacts on city climates and the urban environments. In megacities, such as Kolkata, Delhi, Mumbai, or Manila, motorized transport (excluding public transport) accounts for 60–70% of the total traffic.

2.3. Infrastructure and urban density

Urbanization has significant implications for urban energy demand by affecting the urban infrastructure and building stock. The concentration of economic activity results in a clustering in the inner city and a resulting scarcity of land and the necessity to erect multi-level buildings (Parikh & Shukla, 1995). Such a development has both direct and indirect impacts on energy consumption. The direct impacts on energy consumption are not as large as the indirect ones, the latter being related to the production of the materials required, such as cement or steel. Growing cities show great demand for energy-intensive products. In general, the construction and development of the urban infrastructure, including the construction of roads, bridges, office buildings, sewage networks, power plants, etc., is associated with a high-energy input (Jones, 2004). Also, usage and maintenance of the infrastructure, such as sewage networks, lighting, or water and waste treatment facilities, require additional energy.

A distinctive feature of compact and dense cities is the urban heat island (UHI) effect. Sealed surfaces, such as roads, buildings, and other constructed surface areas, absorb and retain solar irradiation. The displacement of trees reduces natural cooling effects of shading and evapotranspiration. As a consequence, the UHI effect increases the city air temperature by 1–3 °C relative to the surrounding area. Furthermore, waste heat of space heating and cooling, and traffic increase the UHI effect further (Akbari & Konopacki, 2005; Ewing & Rong, 2008; Rosenfeld et al., 1995). Fig. 3 illustrates the issue.

Plantings, public green spaces, and open spaces can mitigate the UHI effect. In terms of energy consumption, on the one hand, the UHI effect enables heat energy savings, because of a higher city air

Table 4
Modal shares of urban transport trips in Asian megacities in 2005 (%).

City region	Transport mode					
	Walking	Non-motorized vehicles	Paratransit	Public transit	Motorcycles	Private cars
Tokyo	8	0	0	53	17	22
Bangkok	1	5	5	40	17	32
Jakarta	23	2	3	25	13	34
Manila	12	3	39	13	3	30
Beijing	12	48	6	20	2	12
Kolkata	15	9	40	6	10	20
Delhi	20	12	53	8	0	7
Dhaka	40	20	8	20	4	8
Mumbai	15	3	28	9	20	25

Source: Laquian (2005), own illustration.

Notes: Non-motorized vehicles include bicycles, rickshaws, and pedal-powered betjaks. Paratransit includes motorized vehicles such as 'aby taxis' tempos, jeepneys, autorickshaws, motorized tricycles, helijaks tuktuks, samlor, and various kinds of two-, three- or four-wheeled vehicles. Public transit includes buses, trams, heavy rail transit, light rail transit, subways, commuter rail, and bus rapid transit systems. Motorcycles include motorized two-wheel private vehicles. Private cars include taxis, rental cars, and limousines.

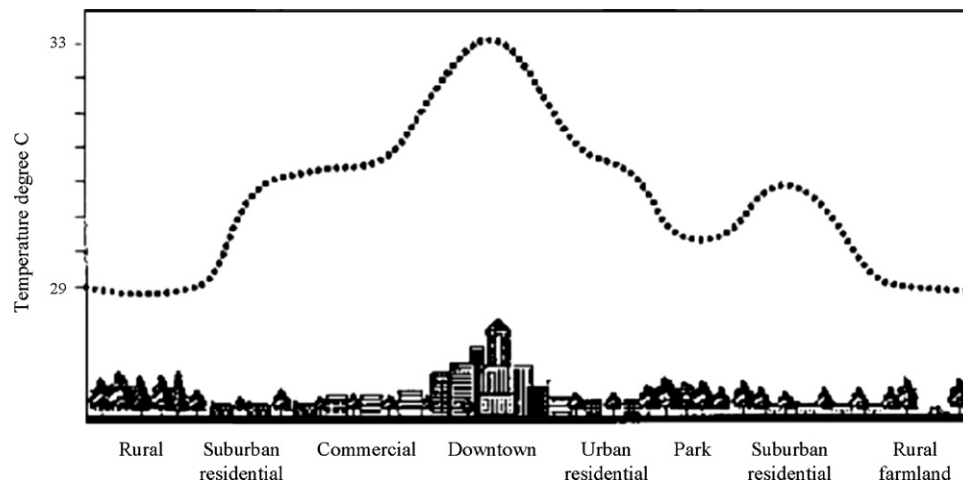


Fig. 3. A general urban heat island profile.

Source: Giridharan, Ganesan, and Lau (2004), modified.

temperature while, on the other hand, it leads to increasing energy consumption due to cooling and the use of air conditioners (Ewing & Rong, 2008). The magnitude of the UHI effect mainly depends on the size and density of the city. Overall, UHIs result in a thermal degradation of the city climate, and in increasing emissions.

Although dense and compact cities (with respect to urban structures) show additional energy consumption due to UHIs, a high site density has advantages compared to less compact areas. For instance, high density cities, such as Hong Kong, show far lower transport energy consumption than low density cities, such as Houston (in this example by a factor of 18). Furthermore, comparing US-American cities with European cities, European cities have a five times higher site density, but US cities consume 3.6 times as much transport energy per capita on average (Karathodorou, Graham, & Noland, 2010; Steemers, 2003). Especially in US-American cities, the process of urban sprawl leads to a more or less unbounded city growth into the city's surrounding area. Scattered settlement fragments in suburban areas demand a higher transport and supply effort due to lower densities. Moreover, buildings in densely-built areas compared to detached houses in suburban areas have fewer exposed walls that release waste heat. Hence, cities characterized by urban sprawl tend to be less energy-efficient (Girardet, 1999; Park & Andrews, 2004). Important issues with regard to the energy demand of buildings are energy efficiency measures, such as improvement of thermal insulation or the installation of modern heating systems. Since urbanization has reached saturation in most developed countries, suburbanization and urban sprawl (and with it energy-related changes within the building stock and urban structures) are major tasks in order to mitigate increasing energy consumption. In the United States, as of 1997, single-family detached housing, with a share of 73.4%, accounted by far for the largest share of residential energy consumption (Lovaas, 2004).

Furthermore, urbanization increases the importance of the informal building sector, especially in developing countries. Rapidly growing cities suffer from a decreasing degree of manageability, and with it an uncontrolled diffusion of illegal housing. More specifically, urban planning in developing countries can often not keep pace with the growth rates of most cities (Balbo, 1993).

Table 5 shows population shares in illegal housing as an illustration. Within most cities in developing countries, illegal housing and informal settlements play a significant role. Even in cities, such as Mexico City and Sao Paulo, which rather can be assigned to more developed regions, vast shares of the population live in illegal housing. An increasing share of illegal housing and unplanned city growth leads to an increasing share of urban fragmentation. More-

over, fragmentation certainly raises the costs of urbanization due to spatial discontinuity, such as inefficient land use (Balbo, 1993). For example, to supply planned residential areas with water or electricity, infrastructure has to pass through illegal settlements without serving them. Associated with it is the risk of illegal hook-ups and connections to the water or electricity network and so increasing costs (Balbo, 1993). Furthermore, we can assume that, because of lacking basic energy infrastructure, the usage of traditional energy sources is widespread. Hence a measurement of the actual energy consumption in those cities is very difficult.

2.4. Private households

Urbanization changes consumer needs and the lifestyles of private households. In particular, changes in consumer needs and behavior especially affect urban energy demand. Generally speaking, an urban population is more dependent on commercial products and services than a rural population (Clancy, Maduka, & Lumampoa, 2008). Whereas rural households are able to cover a certain amount of commodities by in-house production, urban households tend to purchase products and services from commercial production. Commonly, commercial production requires more energy input than in-house production in rural households does (Jones, 1989). Moreover, the transfer of formerly domestic production activities on the market changes the type of the energy source used, with an increasing usage of modern energy sources instead of traditional ones. In line with urbanization, economic development basically affects consumer behavior. The driver of increasing energy consumption is not only a growing population, but rather

Table 5
Population shares in illegal housing.

City	Population (millions)	Population in illegal housing (%)
Jakarta	8	62
Manila	5.6	40
Dehli	7.5	40
Karachi	8	50
Addis Ababa	1.6	85
Cairo	5.7	54
Mexico City	16	50
Sao Paulo	13	32
Bogotá	5.5	59
Caracas	3	34

Source: Pacione (2009), own illustration.

increasing per capita consumption and changing consumer needs and behavior and lifestyles (Satterthwaite, 2009).

Usually, urbanization accompanied by increasing incomes leads to a change in consumer needs, which results in an increasing energy consumption, e.g., due to a growing number of electrical appliances in urban households. In order to mitigate increasing energy consumption of households, policy-makers in developed countries try to enforce energy efficiency improvements, e.g., for electrical appliances. Associated with attempts to raise energy efficiency is the occurrence of the “rebound effect” (Berkhout, Muskens, & Velthuisen, 2000; Brännlund, Ghalwash, & Nordström, 2007; Madlener & Alcott, 2009). With increased energy efficiency and thus reduced energy costs, people tend to consume more goods (and more energy) due to a rise in real disposable income. For example, if gas costs less per transport unit, automobile use (or mileage driven) may increase. Thus, this partially offsets the energy-saving potential expected from engineering-economic calculations alone.

In developing countries, a shift of household behavior and consumption patterns under conditions of urbanization is often not clearly visible. Life in informal slum settlements often does not allow for the adoption of urban lifestyles or behavior. Due to lacking basic services, such as electricity access, urban life of the poor resembles rural conditions.

2.5. Urbanization effects

Various effects and mechanisms of the urbanization process show substantial impacts on urban structures and energy consumption. Fig. 4 summarizes key effects that emerge under conditions of urbanization. It also shows related issues on the global and local level, respectively.

Urbanization and economic development accompany each other and both affect urban structures. The relationship varies with regard to differing spatial and temporal contexts. While economic development can lead to increasing urbanization, high urbanization rates can generate economic growth (Pacione, 2009). During the phase of industrialization in Northern America and Europe in the 19th and 20th century, economic development was inextricably linked with urbanization. Today, with regard to the situations prevalent in many developing countries, an increasing imbalance between these processes becomes obvious.

Apart from impacts on the local urban level, economic development and urbanization also affect global issues, such as climate change and increasing scarcity of resources. Technological innovations, especially concerning innovative energy technologies, play an important role in order to mitigate climate change and the scarcity of resources, and also to foster sustainable development. Both in developed and developing countries, the achievement of sustainability has been at the top of the political agenda since the 1980s. In view of the fact that the urban population will have doubled within the next 40 years in most African and Asian developing countries, and given that the number of emerging megacities is increasing rapidly, sustainable urban development will become a future challenge of paramount importance.

The cross-country analyses of Jones (1989, 1991, 2004) and Parikh and Shukla (1995) estimated the impact of urbanization on urban energy demand. Both studies used aggregated datasets for groups of countries, and conducted statistical analyses in order to measure the relationship between urbanization, various other parameters (such as industrialization and urban density), and energy demand. Despite some data differences, the two studies largely confirm each other. In Jones' study, the urbanization elasticity of modern energy per \$ of GDP, i.e. the percentage change in energy consumption per unit of GDP when urbanization changes by 1%, is 0.48 (per capita 0.45). The urbanization elasticity of traditional energy consumption per \$ of GDP and per capita is statistically not different from zero. Jones' urbanization elasticity of total energy per unit of GDP is 0.35 (per capita 0.30). Parikh and Shukla (1995) only consider the urbanization elasticity of total energy per capita, amounting to 0.47, which is comparable to Jones' elasticity estimate for modern energy per \$ of GDP (0.48).

Table 6 shows future energy demand calculations according to urbanization growth rates and elasticity estimates of Jones' (1989, 1991, 2004) and Parikh and Shukla's (1995). Because of high urbanization rates in less developed regions in the next decades, energy demand could increase by about 6.5–10.4%, depending on the different elasticities estimated. According to this, the least developed regions could increase their energy demand by about 7.8–12.5%. Regarding the world regions, future energy demand will increase significantly in Africa (6.6–10.5%) and Asia (7.1–11.4%) due to urban growth. Given these estimates, effects and mechanisms of urban-

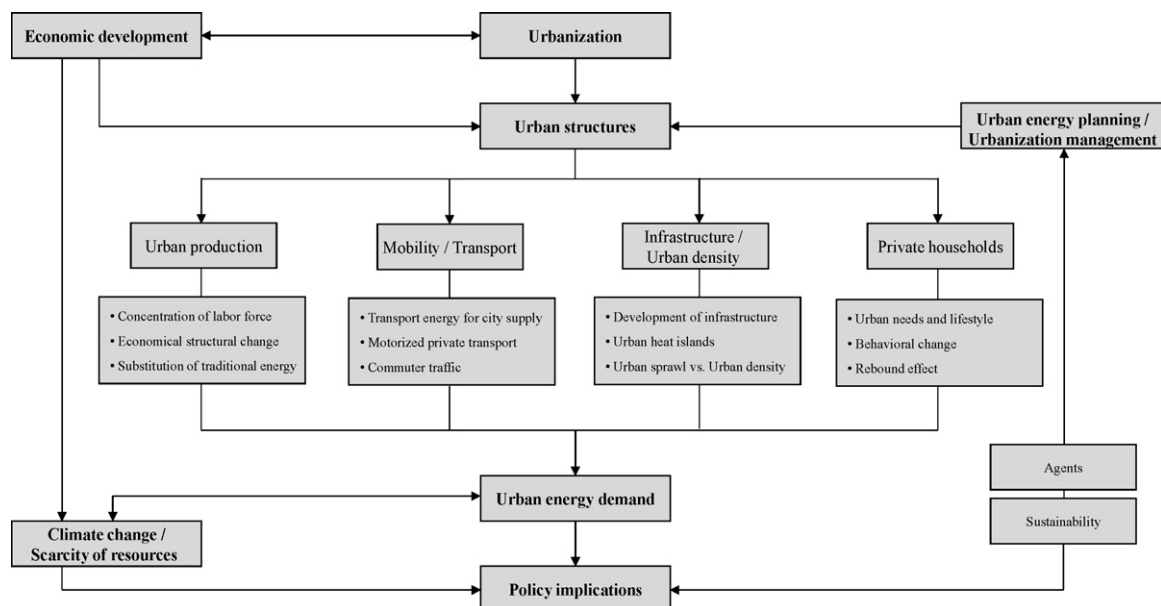


Fig. 4. Impacts of urbanization on urban structures and energy demand.

Source: own illustration.

Table 6

Future energy demand according to urbanization elasticities and urbanization prospects.

Urbanization elasticity	Urban growth 2010–2050 (percentage)	Jones				Parikh and Shukla
		Modern energy		Total energy		Total energy
		Per \$ dollar of GDP 0.48	Per capita 0.45	Per \$ dollar of GDP 0.35	Per capita 0.30	Per capita 0.47
World	19	9.1	8.6	6.7	5.7	8.9
More developed regions	11	5.3	5.0	3.9	3.3	5.2
Less developed regions	21.7	10.4	9.8	7.6	6.5	10.2
Least developed countries	26.1	12.5	11.7	9.1	7.8	12.3
Africa	21.9	10.5	9.9	7.7	6.6	10.3
Asia	23.7	11.4	10.7	8.3	7.1	11.1
Latin America/Caribbean	9.3	4.5	4.1	3.3	2.8	4.4
Northern America	8.1	3.9	3.6	2.8	2.4	3.8
Europe	11.2	5.4	5.0	3.9	3.4	5.3
Oceania	5.8	2.8	2.6	2.0	1.7	2.7

Source: UNPD (2007), Jones (1989), Parikh and Shukla (1995), own calculations and illustration.

ization have major impacts both on energy demand and related pollutant emissions.

3. Relevance for developed and developing countries

The relevance and consequences of the effects of urbanization considered vary in particular cases. Specifically, the effects differ in a spatial and temporal context as well as between developed and developing countries. In most developed countries, the majority of the population lives in cities, so that urban growth has often almost reached saturation. Urbanization has been accompanied by industrialization that has also led, apart from a quantitative growth of cities, to a qualitative growth. Thus, most developed countries have recourse to a wide range of instruments in order to manage urbanization and mitigate increasing energy consumption in cities. Nevertheless, developed countries face a plethora of problems and challenges in order to achieve a sustainable urban future.

The requirements for achieving sustainable urban development are diverse. The main tasks for urban planners are (1) to ensure the compactness of urban structures and (2) to target a functional mix of urban quarters (Hall & Pfeiffer, 2000). Another important assignment of urban planning should be the prevention of unbounded urban sprawl especially with regard to energy consumption. Of course, developing countries are facing comparable problems and challenges concerning urban (energy) planning, but preconditions and preferences may differ considerably. As already mentioned, developing countries show high urbanization rates accompanied by an increasing number of emerging megacities. In many developing countries, urbanization is not accompanied by economic development. This implies that most of the emerging megacities are characterized by quantitative growth rather than qualitative growth. The results of this imbalance are growing slums and informal settlements, and increasing importance of the informal economy. Developing countries face the task of ensuring a reliable access to basic services, such as water or electricity, and at the same time they have to strive for sustainable urban development. Due to different problems and prior tasks, such as covering basic needs, economic growth, establishment of industries, lacking urban planning, or a lack of environmental awareness, urban sustainable development cannot be implemented along the lines of European or Northern American cities. Hence efficient urban energy planning requires an integrative and multi-disciplinary approach.

4. Policy implications

Given considerable differences with regard to urbanization and phases of economic development in developed and developing

countries, policy implications have been adapted to different conditions and situations in order to manage future energy challenges.

4.1. Developed countries perspective

City planners in developed countries today face major challenges with regard to increasing urban energy consumption, fossil fuel dependency, scarcity of energy resources, and not least climate change. In order to achieve urban sustainability, policy agendas in particular feature energy efficiency measures, the increase of the share of renewable energy sources, the reduction of fossil fuel consumption, and the achievement of decentralized power generation.

The implementation of sustainability measures on the urban scale is mainly characterized by the attempt to achieve a sustainable urban design, which predominantly addresses the extent and degree of urban density and compactness (Lehmann, 2008). The idea of the compact city model goes back to Newman and Kenworthy, and includes characteristics such as high density, defined boundaries containing city growth, mixed land uses, and heavy reliance on public transport (Lehmann, 2008; Newman & Kenworthy, 1989). Compact cities are likely to have advantages compared to sprawling cities with regard to energy consumption, but the issue of density and compactness of cities is more complex. The question remains: to what extent is compactness beneficial with regard to the mentioned urbanization effects, such as UHIs or scarcity of land?

According to Pacione, sustainability measures have to be implemented at different city levels, such as the neighborhood level, the district level, and the metropolitan level (Pacione, 2009). The neighborhood level contains microscale measures. In particular, these measures are related to urban design. A major challenge within the design of urban structures in developed countries addresses the building stock. Based on the example of energy use in London shown in Fig. 5, it becomes obvious that major shares of energy consumption are attributable to the building stock.

The building stock (domestic and commercial buildings) accounts for 61% of total energy consumption. Energy savings require the development and usage of energy-efficient appliances (especially heating and cooling systems) and retrofitting of the existing building stock. Additional measures related to urban design and architecture should be the fostering of town houses, low-rise flats or passive solar design in order to reduce energy consumption. Passive solar building design aims at optimal solar gain and indoor climate by planned siting, orientation, layout, and landscaping, in order to reduce the need for space heating and cooling (Pacione, 2009; Tombazis & Preuss, 2001).

At the district level, the adoption of Combined Heat and Power (CHP) plants should be supported in order to achieve a more decen-

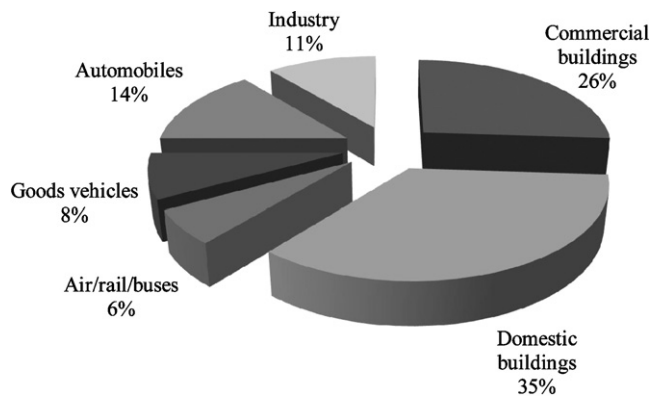


Fig. 5. Energy use breakdown for London 2000.

Source: Steemers (2003), own illustration.

tralized energy supply. In doing so, the adoption of small-scale CHP units for the energy supply of individual buildings or groups of buildings can contribute to achieving this goal. CHP units linked to district heating systems are widespread in Scandinavian countries. In Helsinki, for example, CHP units linked to district heating already account for 81% of the heat and 72% of electricity used (Pacione, 2009).

At the metropolitan level, the most important challenge is to achieve sustainable mobility and transport. Associated with high degrees of urbanization are the increasing numbers of road vehicles and a high reliance on private transport, especially private automobile use. For instance, due to the convenience of private automobile use or developments in retailing, a major challenge for city planners is to shift automobile users to other transport modes (Pacione, 2009).

In many OECD countries, motorized private transport causes up to 50% of atmospheric lead emissions in urban areas, where congestion costs are major factors: in the city center of Athens, road traffic moves at an average of only 4–5 mph (7–8 km/h), in Paris 11 mph (18 km/h) and in London 12.5 mph (19 km/h) (Haughton & Hunter, 1994). In order to reduce energy consumption and emissions, new drive technologies for automobiles, such as electrical drive engineering, fuel cells, or hybrid e-vehicles, have to be developed, supported, and adopted.

4.2. Developing countries perspective

In order to achieve a sustainable urban development a multi-dimensional coordination of local, national, and global agents and responsibilities is necessary. Furthermore, urban energy planning is inextricably linked with urban planning. In the case of developing countries, urban planning has to be multi-dimensional, participatory, and multi-sectoral (Hall & Pfeiffer, 2000; Todoc, 2008).

In a global context, financial and technical development aid has to be intensified in order to enable developing countries to achieve a sustainable economic development and to contribute to the protection of the global climate. Poverty and climate protection are closely linked to each other. Investments in climate protection and reliable and clean energy technologies in developing countries can help to improve the quality of life and reduce poverty.

On the national level, administrative requirements have to be complied with. In order to achieve sustainable development, principles of good governance – such as democratic legitimation, participation, constitutionality, human rights, political transparency and efficiency – have to be fulfilled (Wohlmuth, 1998). Good governance represents both the precondition and the instrument of sustainable development. In this context, a comprehensive deconcentration of responsibility and political power is

necessary. Local decision-makers need more power of decision in order to manage urbanization.

On the local level, a close cooperation between decision-makers and society is crucial. Urban agents, such as neighborhood communities, NGOs, religious groups, or self-help organizations, have to be included in the decision process. This also includes involving the segments of population that live and work in the informal sector (Hall & Pfeiffer, 2000). Further basic requirements on the local level are the creation of legal certainty and the warranty of accountability and ownership (Fiedler, 2009).

Given rapidly growing cities and an increasing importance of the informal economy, city planning in developing countries has to be understood as management of urbanization. Urban management must be integrative and has to consider all city sectors, also including energy planning.

Various basic measures should be adopted in order to manage urbanization. With regard to high shares of the informal sector, conception and formalization, e.g. of property, should be a first step in order to formalize informal economic activities. This establishes a basis that allows the raising of taxes (Fiedler, 2009). The clarification of legality and the generation of new sources of revenues provide the chance to develop urban basic infrastructure, such as water or energy supplies. According to calculations of the United Nations, the supply with electricity and reliable energy sources for 50 million people costs about US\$ 1.4 billion (UN-Habitat, 2006), or US\$ 28 per capita.

A sustainable urban energy supply system includes the usage of renewable energies, such as biomass, solar, wind, or hydro. Also, biogenic wastes can be used as a substrate in biogas plants. With regard to the problems of waste management in many cities, this implies a double dividend (UN-Habitat, 2006). Despite the supply-related problems of high population and building density in informal settlements, high densities offer a unique opportunity to provide electricity. According to the calculations of UN-Habitat (2006) for Nairobi, legal, metered electricity connections for two million people living in slums can cost as little as US\$ 200 million in terms of initial investment. This would enable to deliver electricity to a household for as little as about US\$ 1.25 per month (UN-Habitat, 2006).

Another measure is the higher taxation of motor vehicles or a city toll for urban areas (Fiedler, 2009). This can be an important instrument in order to reduce private transportation and mobility-related energy consumption and emissions. In addition, this can keep important city space free in order to establish public transportation systems. Apart from establishing public transport systems, the additional revenues can be used to develop the urban infrastructure. Moreover, urban logistic and supply systems can be organized more efficiently.

By providing city space for informal settlements and by creating a legal framework, the dynamics of city growth can be organized to some extent. In order to generate housing space, a micro-financing system can be useful (Fiedler, 2009).

A further approach within the management of urbanization is area zoning. Zoning can be understood as a pre-stage of a land development plan that allows the efficient location of public facilities or companies (Fiedler, 2009). Zoning is necessary in order to achieve a compact city structure and a functional mix of urban quarters. This implies positive effects with regard to traffic volume.

5. Conclusion

In this paper, we have discussed that future urban growth will mainly occur in less and especially least developed countries, while urbanization has almost reached saturation in more developed countries. Associated with the process of urbanization is increas-

ing energy consumption due to various effects and mechanisms that influence urban structures and human behavior. The relevance and impacts of these effects differ considerably between more and less developed countries. Due to differences in city structures, economic development, and administrative structures, urban policy implications and management of urbanization have to be adapted to the different urban conditions and situations. In developed countries, urban energy planning concerns different city levels. Prior challenges address energy efficiency measures and the increase of the share of renewable energy sources. Especially the current design (e.g. building stock) of cities is responsible for high urban energy consumption. In order to manage urbanization and to mitigate increasing energy demand in developing countries, particularly with regard to emerging megacities and the tendency of 'overurbanization', policy measures have to be multi-dimensional, participatory, and multi-sectoral. Major challenges are the formalization of the informal economy, the development of basic infrastructure, and the integration of energy planning in urbanization management.

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